



Spine Anatomy and Clinical Correlations of the Thoracic Lumbar and Lumbar Region

Dr. Kyaw Lin Fall 2025

Special Thanks to DR. Zusman

Objectives

► Thoraco-Lumbar Region

1. Infections and Malignancy:

- Understand the presentation and management of infectious diseases, rheumatologic conditions, and malignancies involving the spine as mentioned in the lecture material.

2. Scoliosis:

- Be familiar with scoliosis and structural changes of the axial spine especially involving the thoracolumbar region.
- Be able to explain the measurement of Cobb angle.
- Understand the conservative and surgical management of scoliosis as defined by Cobb angle.
- Differentiate between pediatric and adult scoliosis and discuss treatment options.

► Lumbar Region:

1. Lumbar Pathologies:

- Differentiate the etiology of lumbar spine pathologies including spinal stenosis, spondylosis, and spondylolisthesis, radicular

2. Discogenic Back Pain and Herniation:

- Recognize and treat discogenic back pain and lumbar herniation.

3. Management of Chronic Low Back Pain:

- Review primary care strategies and osteopathic principles for chronic low back pain.

Definitions

- **Axial pain**-pain in low back
- **Radiculopathy**-symptoms and pain along nerve root distribution
- **Referred pain**-pain perceived in a body part but arising from another area
- **Claudication**-leg pain occurring with ambulation
- **Spinal stenosis**-narrowing of spinal canal
- **Spondylosis**- overall degeneration of the spine.
- **Spondylolisthesis**- displacement of a vertebra in which the bone slides out of its proper position onto the bone below it
- **Disc herniation**-expulsion of disc material
- **Instability**-loss of structural integrity so motion occurs outside normal physiological limits causing or risk of causing pain or neurological effect
- **Cauda equina syndrome**-loss of bowel and bladder function and or progressive or severe neurological loss due to compression of lumbar nerves
- **Neuromodulation**: a therapeutic technique that uses electrical or chemical stimulation to alter nerve activity in the central, peripheral, or autonomic nervous system.

Conditions covered today

Thoraco-Lumbar Region

- ▶ Ankylosing Spondylitis *
- ▶ Metastatic Cancer**
- ▶ Infection (Osteomyelitis, etc.) **
- ▶ Osteoporotic Compression Fracture**
- ▶ Scoliosis: Adult and Pediatric

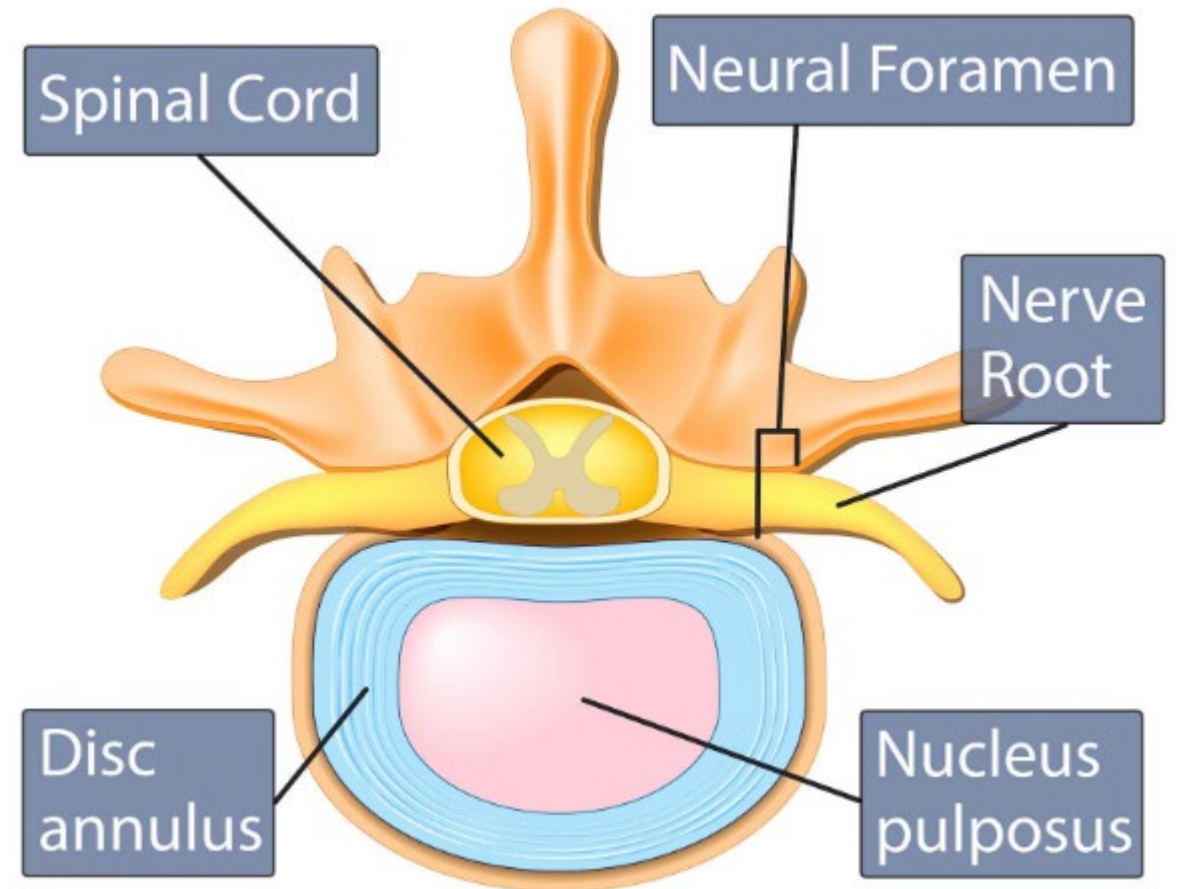
Lumbar Region

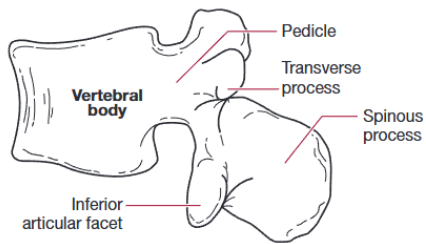
- ▶ Lumbar Spondylosis/Degenerative Disease, Facet Arthropathy
- ▶ Lumbar Spine Stenosis
- ▶ Lumbar Spondylolisthesis
- ▶ Lumbar Disk Herniation

Supplemental topics

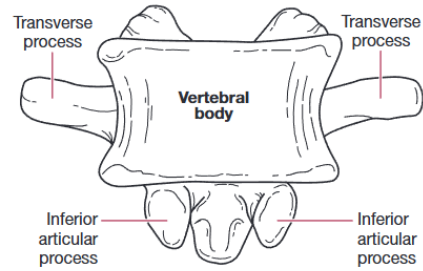
- Cauda Equina Syndrome
- Failed Back Surgery Syndrome

ANATOMY

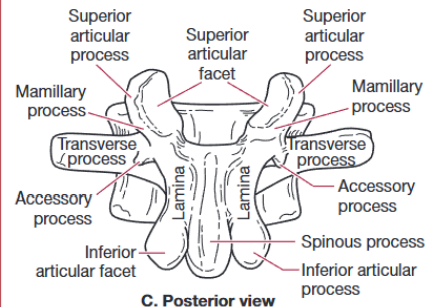




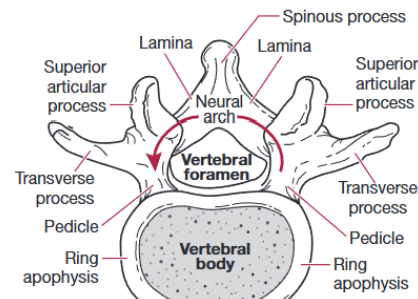
A. Left lateral view



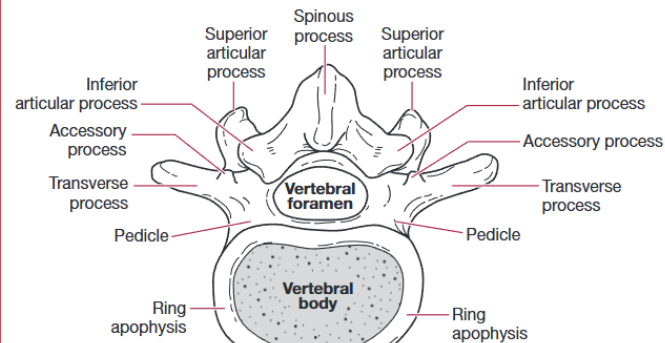
B. Anterior view



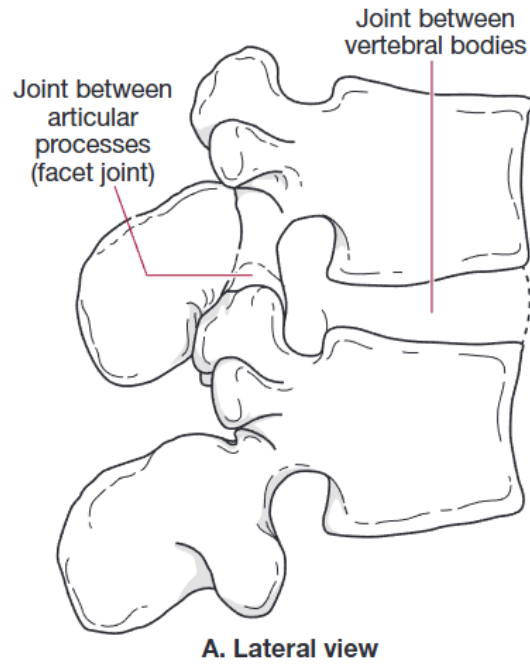
C. Posterior view



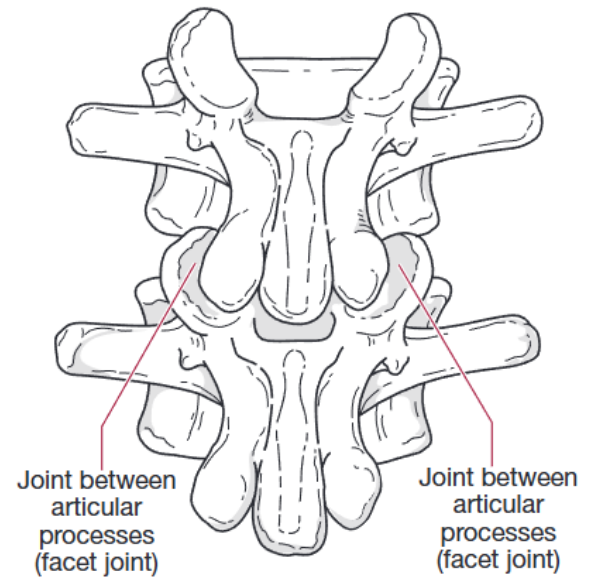
D. Top view



E. Bottom view



A. Lateral view



B. Posterior view

Spinal Load



FIGURE 4-150 Positional disc pressure changes. Relative change in pressure (or load) in the third lumbar disc in various positions in living subjects.

Note: Neutral erect posture is considered 100% in the figures; other postures and activities are calculated in relationship to this.

Source: From Nachemson AL. The lumbar spine: an orthopaedic challenge. *Spine*. 1976;1:59-71.

doi:10.1007/978-93-325-1476-9_100000, 107600000, 00000, associated with a commission



Clinical Conditions By
region

Thoraco-Lumbar Region

Ankylosing Spondylitis



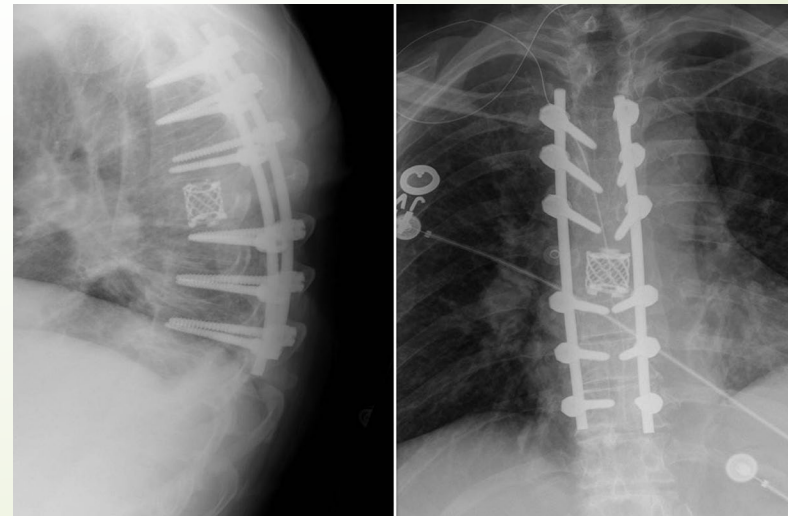
- Young
- >3 months of back pain and stiffness
- Schober test
- HLAB27
- X-ray showing sacroiliitis



Bamboo spine and ankylosed sacroiliac joints.

Neoplasm – Metastatic Disease

- Most spinal Mets – extradural
 - ACS – estimates 1.7 million cases of CA will be diagnosed in 2014
 - 75% metastatic disease – spine, most common osseous location
 - Most common primary cancers
 - Breast
 - Prostate
 - Lung CA
 - Direct decompressive surgery can be considered in addition to radiotherapy
 - Life expectancy at least 3-6 months



Infection

- Discitis
 - Vertebral osteomyelitis
 - **Epidural abscess**
-
- Presentation
 - Pain associated with fever & weight loss
 - Neuro deficits – onset late (week to months after pain)



Excruciating pain localized over spine,
tender to percussion



(3 days)

Radiculopathy



(4.5 days)

Weakness



(24hrs)

Paralysis

*Fever, sweats, or rigors are common, but not always present.

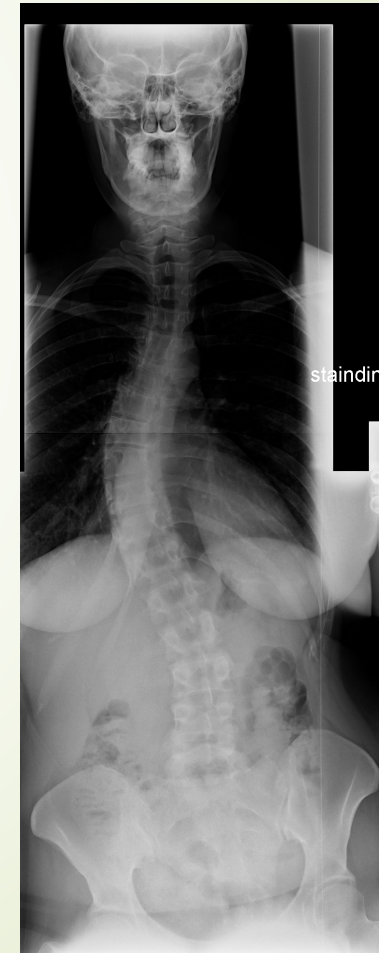
Osteoporotic Compression Fracture

- RF: elderly, postmenopausal women, due to decreased bone density.
- Presents w/ acute, localized back pain following minimal trauma, such as bending or lifting.
- X-rays are typically the first imaging modality.
- Treatment is usually conservative, focusing on pain management, bracing, and osteoporosis medications to prevent future fractures.
- For persistent pain, vertebral augmentation procedures like vertebroplasty or kyphoplasty may be considered.



Scoliosis

- Definition: a lateral curvature of the spine greater than 10 deg. In the coronal plane using the Cobb method
- However, Scoliosis is actually a three-dimensional deviation in the axis of a person's with abnormal curves in the sagittal plane as well as rotationally along the spinal axis.

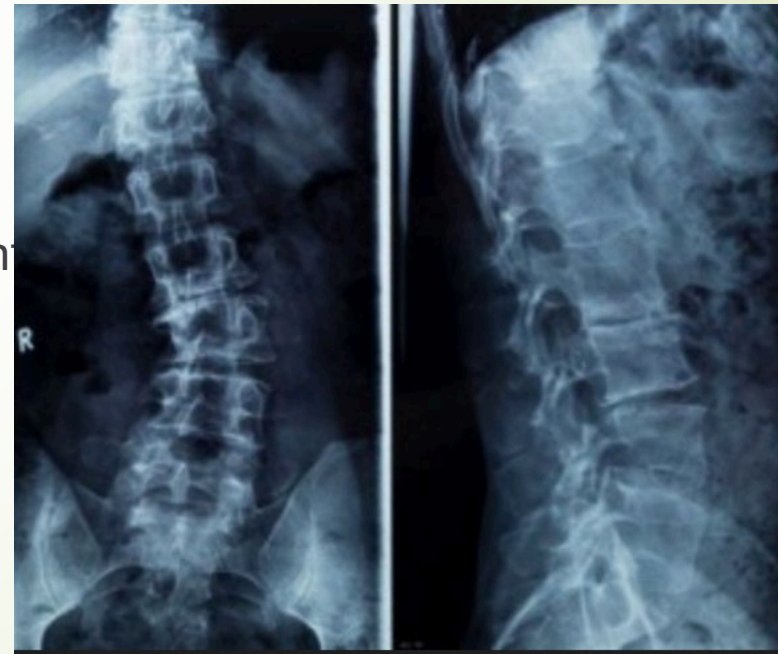




TREATMENT	DEGREE OF ANGULATION (COBB ANGLE)
Observation	<20 degrees
Bracing	20–40 degrees
Surgery	>40 degrees (>35 degrees for neuromuscular diseases)

Scoliosis

- Idiopathic –
 - Childhood/adolescence
 - Etiology unknown
- Congenital –
 - Failure of bony development
- Neuromuscular –
 - CP, MD, polio
- Compensatory –
 - leg-length discrepancy
- Degenerative
 - Adulthood



Adult Scoliosis

➤ Etiology

- Idiopathic: most common
- Antalgic
 - Back pain
 - radiculopathy
- Chronic leg length inequality
- Degenerative spondylosis



➤ Symptoms

➤ Pain

- Pain onset insidious, gradually comes on
- Worse in Morning
- Improved with sitting
 - Unloads posterior facet joints

➤ Postural Changes

- Shorter
- Uneven shoulders or hips

- Neurological symptoms
 - Foraminal nerve compression
 - Central spinal stenosis





Adult Scoliosis Management

- Physical Therapy/Aqua therapy
 - Muscle conditioning and postural exercises
 - Aerobic exercise
- OMM
- Medication – NSAID, acetaminophen
- Bracing generally ineffective
- Surgery

Pediatric Scoliosis

➤ 1. Observation

- **Indication:** Curves <20 degrees or mild curves in young children with ongoing growth.
- **Management:** Regular monitoring every 4-6 months with physical exams and X-rays to assess curve progression.
- **Goal:** Monitor for progression, especially during growth spurts.

➤ 2. Bracing

- **Indication:** Moderate curves (20-40 degrees) in children and adolescents with significant growth remaining.
- **Types of Braces:** ★
 - **Thoracolumbosacral Orthosis (TLSO):** Common for curves in the thoracic and lumbar spine.
 - **Milwaukee Brace:** Typically used for curves higher in the thoracic spine.
- **Duration:** Typically worn 18-23 hours per day; effectiveness depends on compliance.
- **Goal:** Prevent further progression of the curve, not necessarily to correct it.



Pediatric Scoliosis

➤ 3. Physical Therapy

- **Exercises:** Focus on core strengthening, postural alignment, and flexibility.
- **Goal:** Improve function, posture, and possibly reduce pain; not proven to halt curve progression.

➤ 4. Surgical Intervention

- **Indication:** Severe curves (>40-50 degrees) or curves that rapidly progress despite bracing, causing functional or cosmetic issues.
- **Procedure Options:**
 - **Spinal Fusion:** Commonly performed to correct and stabilize the curve; typically for curves ≥ 50 degrees.
 - **Growing Rods:** For young children with severe scoliosis to allow for growth while controlling curve progression.
 - **Vertebral Body Tethering (VBT):** A less invasive option for children still growing, aiming to correct scoliosis while preserving mobility.
- **Goal:** Achieve long-term curve stability, improve alignment, and prevent progression.



Summary:



- Ankylosing Spondylitis: Young men with back pain and stiffness. HLAB27.+ Schober test. X-ray showing sacroiliitis.
- Lumbar Spine Infections: Risk factors: immunosuppressed, Recommend MRI w/contrast. Treat empirically. Mostly *S. aureus*. Consult surgery for deficits especially if epidural abscess.
- Metastatic Cancer: through history and screening of red flags. Maybe from breast, prostate, lung cancers. Recommend MRI w/contrast. Multidisciplinary team-based treatment.
- Adult scoliosis: idiopathic is most common etiology.
- Pediatric scoliosis: 20-40 degrees: bracing effective.



Clinical Conditions By
region

Lumbar Spine



The most common diagnosis
for lumbar spine pain is
Spondylosis or Degenerative
Disease

Lumbar degeneration

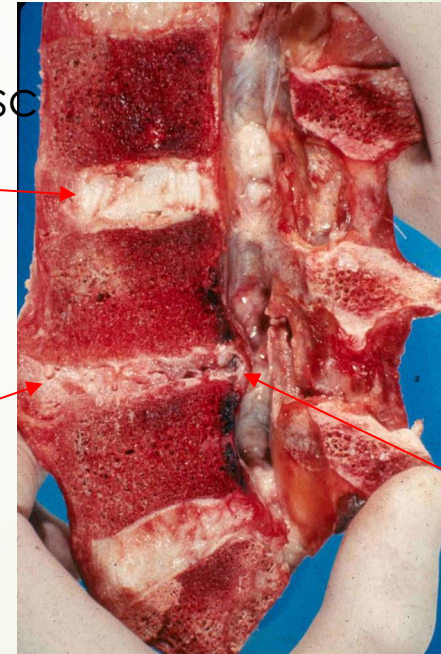
demarcated disc
& anulus



comingled disc
& anulus

normal canal

Stage III
degen.
disc



Disc/osteophyte
central stenosis

3 Stages of Degeneration On MRI



ddback.com

Normal



Stage I



Stage I



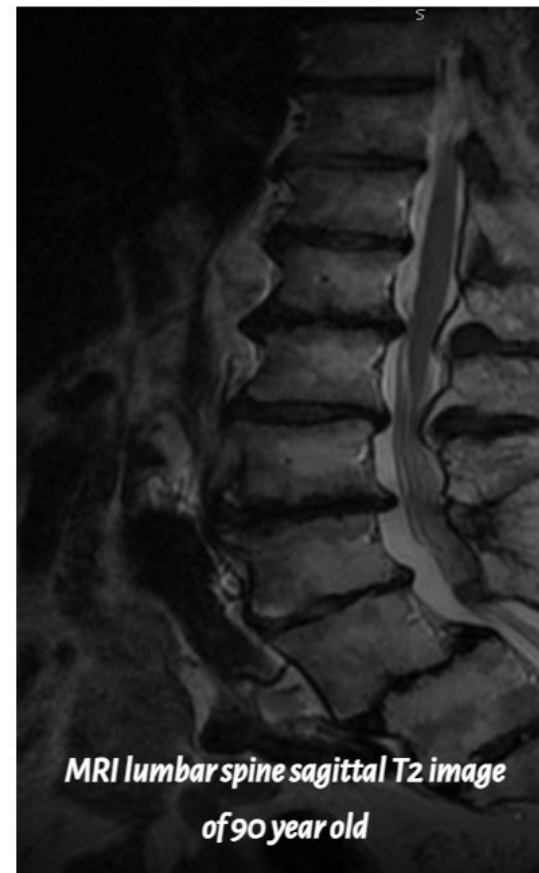
Stage III



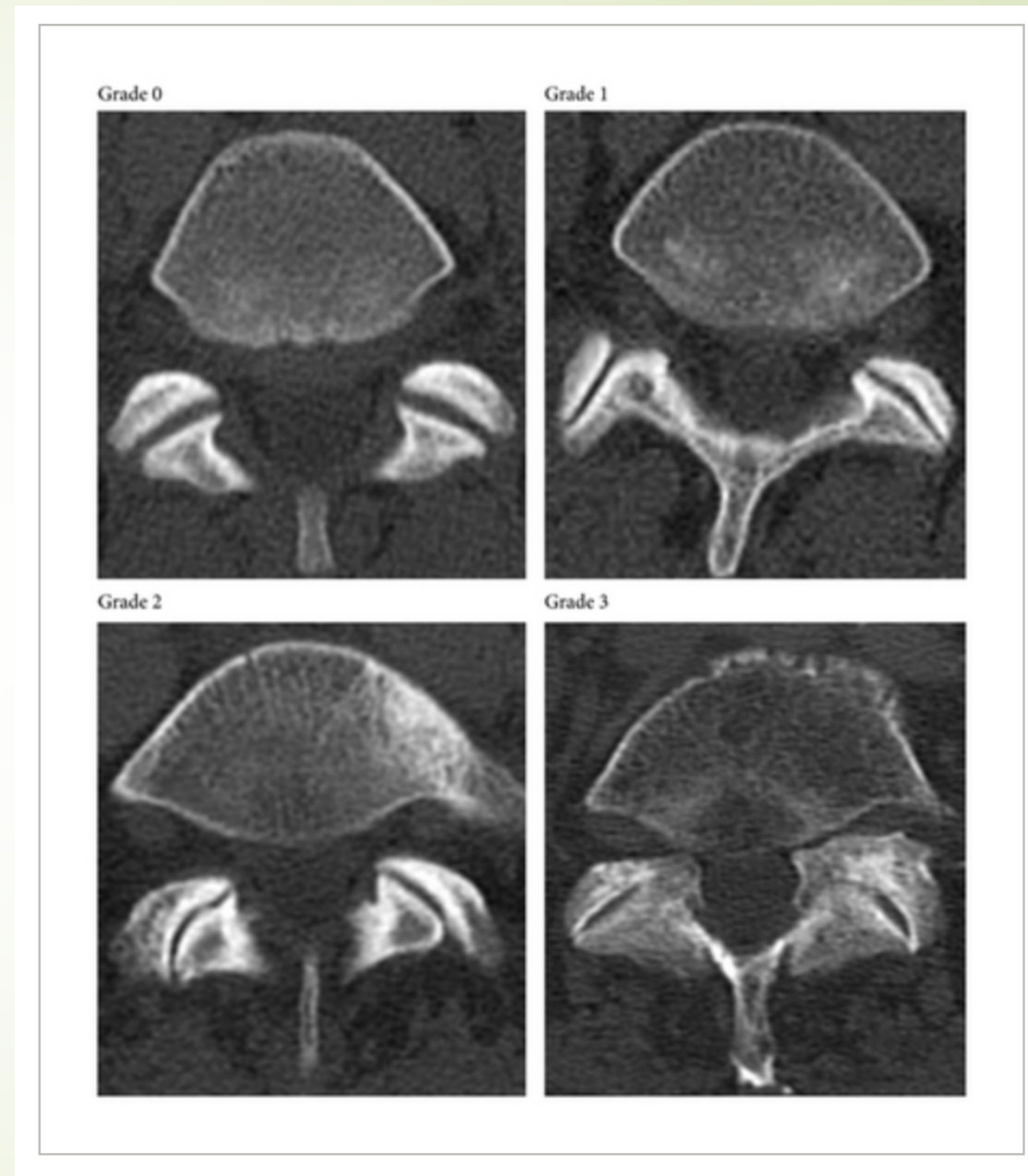
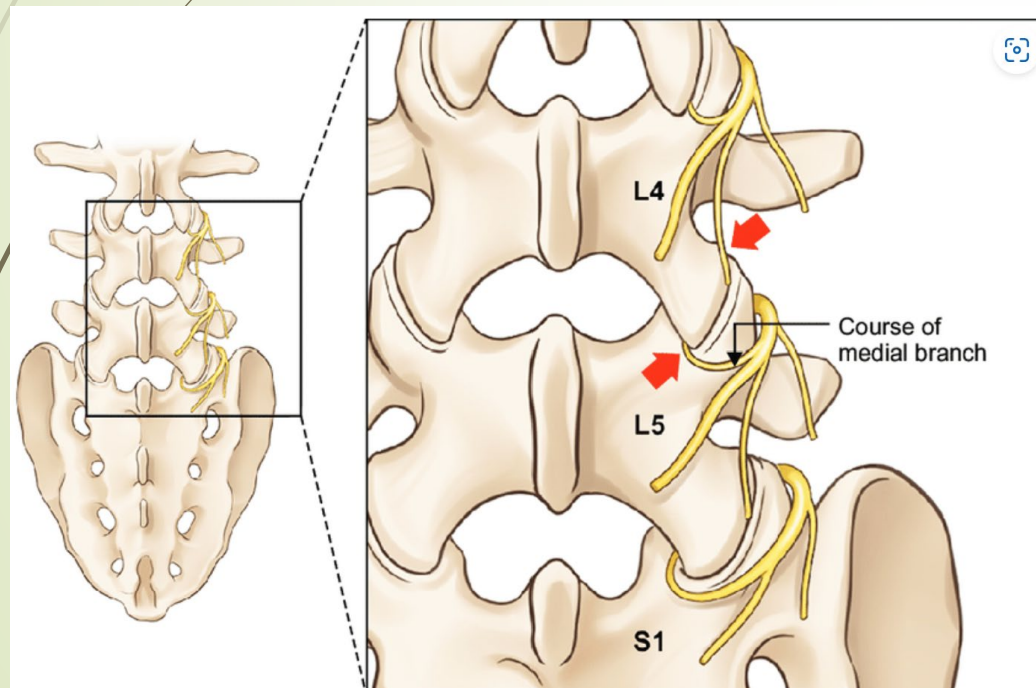
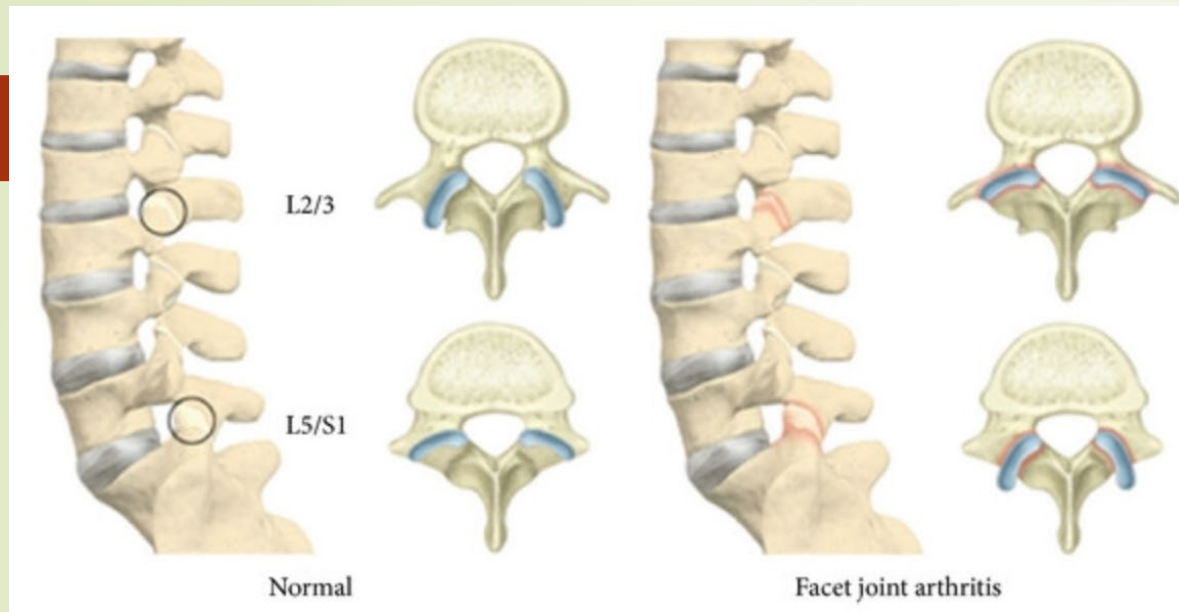
MRI lumbar spine sagittal T2 image
of 30 year old normal)

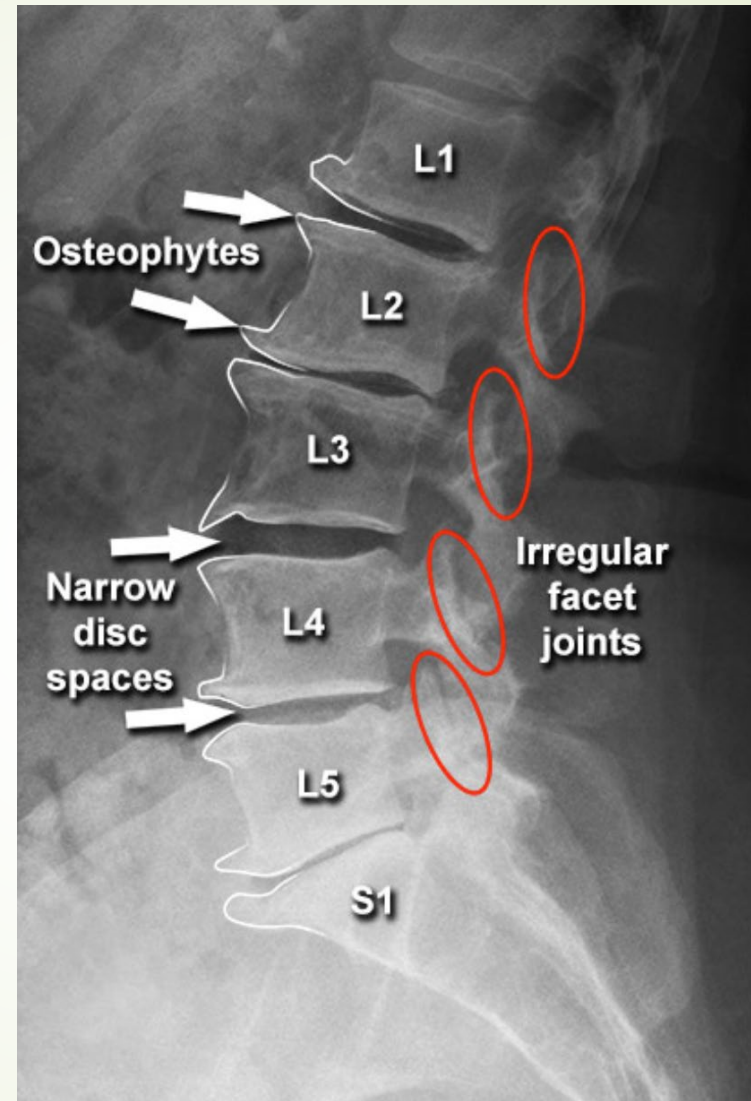


MRI lumbar spine sagittal T2 image
of 70 year old



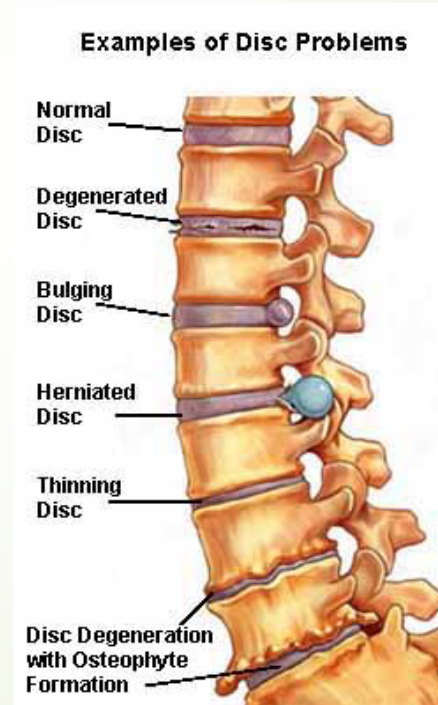
MRI lumbar spine sagittal T2 image
of 90 year old





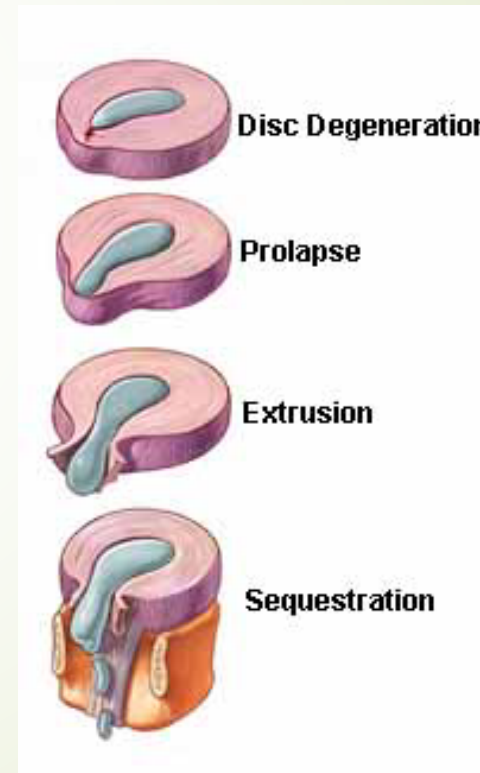
Structural Changes of Disc Degeneration

- Loss of Disc Height
- Loss of Normal Lordosis
- Compromise of the neural canal due to bone stenosis or HNP
- Foraminal compromise
- Instability patterns:
 - spondylolisthesis
 - kyphosis
 - scoliosis



Structural Changes Can Result in Disc Herniation

- Structural integrity reduced with degeneration
- Tears occur in the anulus
- Softer material can protrude thru the tear in anulus
- Displaced material can result in nerve compression





Evaluation of The Patient with spinal disorders



Historical Red flags

When to be concerned Pain is Not Degenerative Disease



- History of diabetes, osteoporosis, cancer, immunosuppression
- Night pain
- Fever or history of infection
- Weight loss
- Progressive weakness
- Bowel or bladder dysfunction
- Widespread neurological symptoms
- Age: less than 20 or greater than 70

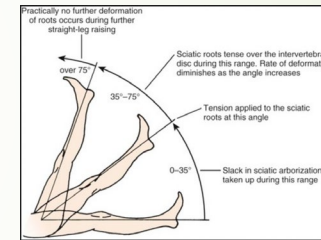
the Clinical Exam-EXAM

- General physical
- Spine specific exam
 - Gait
 - foot drop, widebased, Trendelenberg
 - Inspection
 - Tenderness
 - Palpable vs percussive
 - ROM
 - Neuro changes
 - Nerve tension signs
 - Rectal Exam as indicated by history
 - Psychological screening

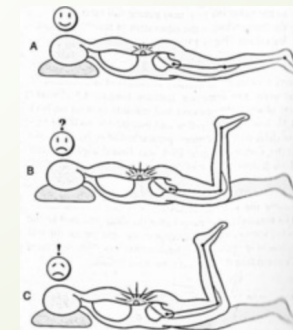


NERVE TENSION Signs

- Increase tension or compression of the compromised nerve root
- Positive when reproduce the symptoms along pattern of nerve root distribution



Sciatic Stretch
L5 or S1



Femoral Stretch
L3 or L4

Importance Clinical correlation of Diagnostic Studies

- Diagnostic Studies help define and localize pathology
- Laboratory
- Electrodiagnostic:
 - EMG/NCV
- Diagnostic radiology
 - Plain Xray
 - Flexion Extension Xray
 - CT
 - MRI

When to order diagnostic studies?



- Neuro deficits Yes
- Severe medical condition
- (+) Red Flags Yes
- Acute non specific back pain No
- Subacute or chronic back pain Depends on Hx
- Persistent radicular pain > 6 weeks Yes
- Claudication suspect Spinal stenosis Yes

How do studies help?

	Soft Tissue	Bone	Nerve	Postop scar	Neoplasm	Trauma	Infection
Xray	poor	Good for gross change & alignment	N/A	N/A	fair	Fair to good	Fair to good
CT	Fair to good	excellent	good	N/A	good	excellent	good
MRI	excellent	good	excellent	Excellent with gadolinium	excellent	Fair to good	excellent



Clinical Conditions By
region

Lumbar Spine

Epidemiology – Low back pain

- Second most common reason to seek medical attention²
 - Lifetime prevalence range from 60-90%⁴
 - Annual incidence is 5%⁴
- 15% of all sick leave from work³
- Most common cause of disability for persons <45 yrs of age³
- 85% of cases no specific diagnosis can be made¹
 - 1-3% will have lumbar disc herniation⁴
- 80-90% will improve within 1 month even without treatment

1. Acute low back problems in adults: assessment and treatment. Agency for Health Care Policy and Research. *Clinical practice guideline. Quick reference guide for clinicians*. Dec 1994(14):iii-iv, 1-25.
2. Cypress BK. Characteristics of physician visits for back symptoms: a national perspective. *American journal of public health*. Apr 1983;73(4):389-395.
3. Cunningham LS, Kelsey JL. Epidemiology of musculoskeletal impairments and associated disability. *American journal of public health*. Jun 1984;74(6):574-579.
4. Frymoyer JW. Back pain and sciatica. *The New England journal of medicine*. Feb 4 1988;318(5):291-300.

Axial or Radicular?



- Axial: in the Back
- **Spondylosis: Facet arthropathy**

- Radicular: along the nerve roots, down the lower extremities.
- **Disc Herniation**

- Mixed symptoms:
- **Spinal Stenosis**
- **Spondylolisthesis**



Evaluation

Back Pain

■ Mechanical (97%)¹

- Degenerative processes
- Congenital disease
- Deformity

■ Non-mechanical (1%)

- Neoplasia
- Infection
- Inflammatory arthritis
- Scheurmann's disease
- Paget's disease of the bone
- Osteoporosis/osteopetrosis

■ Visceral Disease (2%)

- Disease of the pelvic organs
- Renal disease
- Aortic aneurysm
- GI Disease

"Red Flags" should prompt further work-up²

- Age >70 or < 20 yrs
- Systemically ill patients
- Temp > 100° F
- History of Malignancy
- Recent Infection
- IV drug use
- Diabetes
- Immunosuppressed patients
- Urinary tract or spinal surgery
- Recent Trauma
- Pain at rest or pain for > 4 wks
- Unexplained weight loss

¹Deyo RA, Weinstein JN. Low back pain. The New England journal of medicine 2001;344:363-370.

²Greenberg MS, Greenberg MS. *Handbook of Neurosurgery*. 7th ed. Tampa, Fla.: Greenberg Graphics ; 2010.

Clinical Presentation

❖ Axial = back pain

- ❖ Back pain

- ❖ Non radicular leg pain

- ❖ Can have specific physical exam

- ❖ Diagnostic studies may not always be reliable for localization

❖ Radicular = Leg pain

- ❖ Pain in nerve root dermatome

- ❖ Neuro changes

- ❖ +SLR

- ❖ Diagnostic study correlates



Acute Low Back Pain

- Physical findings

- Inspection: atrophy?
- Palpation: tenderness?
- ROM: decreased?

- Diagnostics

- Plain lumbar spine x-ray not indicated for first 6 weeks unless red flags are present,



Chronic Low Back Pain

➤ Considerations

- Pain lasting > 3 months
- Rule out (R/O) tumor (metastatic?), kidney disease, abdominal aneurysm, osteoporotic fracture, etc.
- Explore **psychosocial issues** again, particularly in worker's comp. cases and if litigation is involved



Chronic Low Back Pain



Pain* management center

- Activity Restrictions
- Ergonomic Optimization
- Comprehensive directed rehabilitation
- **OMM**
- Medication management
- Interventional Procedures
- Psychosocial counseling

Clinical case

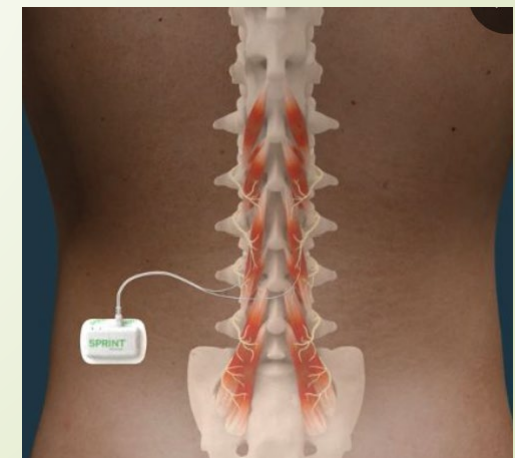
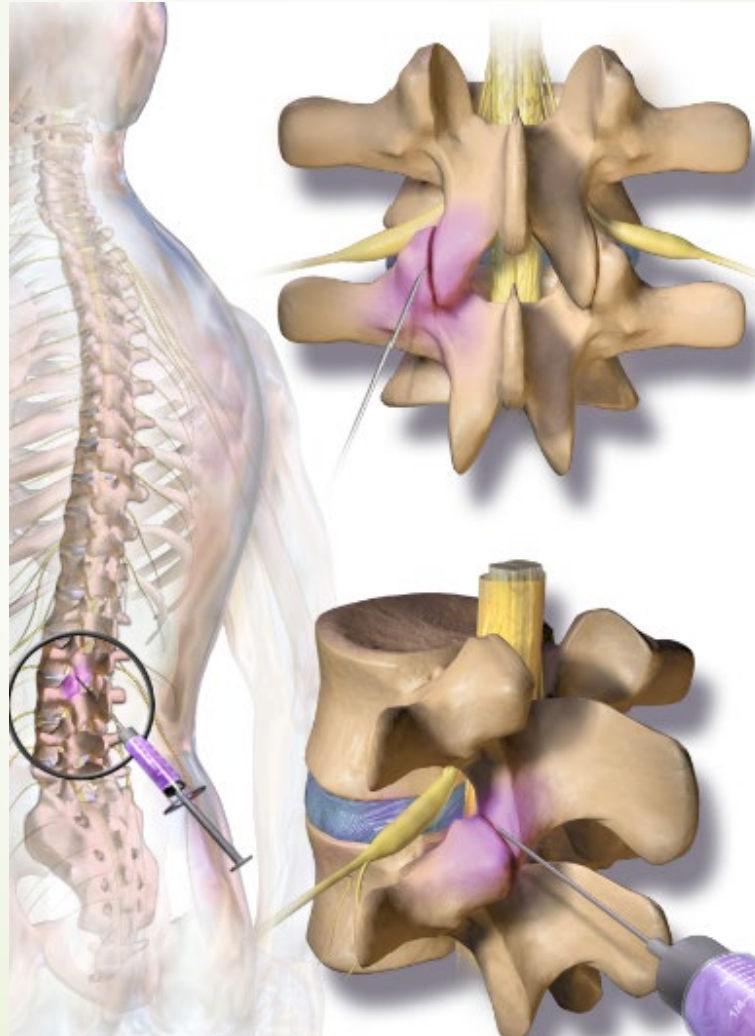
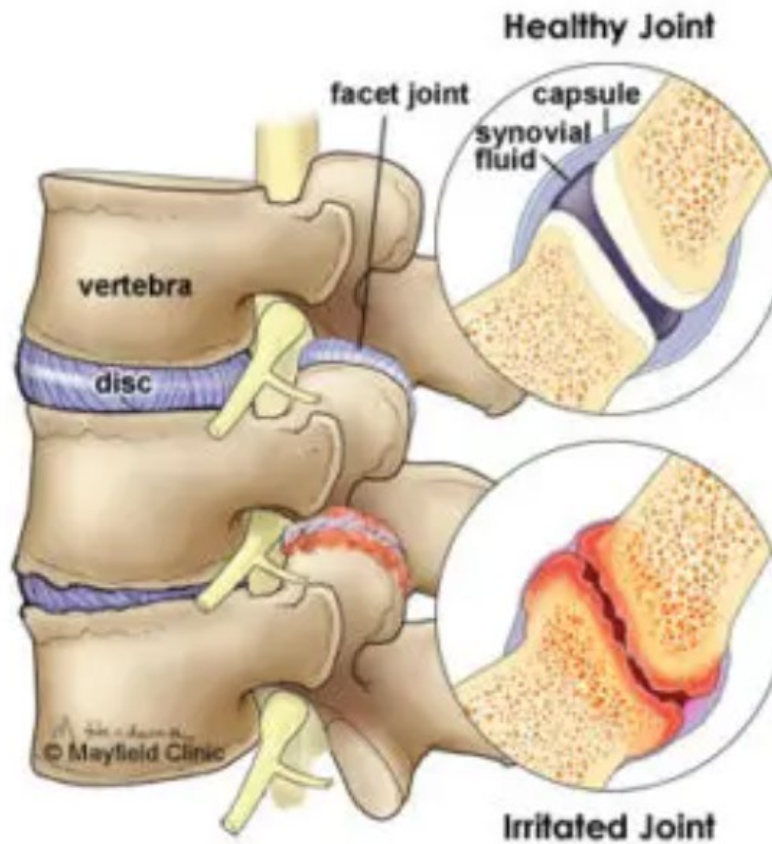
- A 60-year-old woman presents to her primary care clinic with complaints of chronic low back pain that has persisted for several years but has recently worsened over the past three months.
- She describes the pain as a deep, aching sensation located on both sides of her lower back, which occasionally radiates to her buttocks but does not extend down her legs.
- The pain is worse in the morning and after periods of inactivity and is aggravated by twisting or bending backward. She finds that the pain improves somewhat with gentle movement and after taking NSAIDs. Her medical history includes osteoarthritis and a history of heavy lifting due to her previous job as a warehouse worker.
- On examination, there is tenderness over the lumbar paraspinal area, especially near the facet joints. Extension and rotation of the spine reproduce her pain, while straight leg raise and neurological examination of the lower extremities are normal.
- **Question:**
What is the most likely diagnosis, and what are the recommended initial management strategies?

► **Answer:**

The most likely diagnosis is **axial low back pain due to lumbar facet arthritis**.

- Initial management includes patient education on posture and body mechanics, activity modification, modalities, oral medications* for pain relief.
- OMM: counterstrain, ME, HVLA
- Physical therapy focused on core strengthening and lumbar stabilization may also be beneficial.
- If pain persists, referral for interventional pain management, such as facet joint injections or lumbar medial branch block/RFA.
- If the above does not work, consider neuromodulation via peripheral nerve or spinal nerve cord stimulator prior to surgical consult.

Facet Arthropathy



Clinical case

- ▶ A 35-year-old man presents to the clinic with a 2-week history of low back pain radiating down the back of his right leg. He describes a sharp, shooting pain that travels from his lower back to his buttock, thigh, and down to the outer aspect of his calf and foot. The pain is aggravated by prolonged sitting and bending forward and improves somewhat with lying down. He also reports numbness and tingling in the sole and outer side of his right foot.
- ▶ On physical examination, there is a positive straight leg raise test on the right side, reproducing the pain down the leg. Motor examination reveals mild weakness in plantarflexion of the right foot, and there is a decreased Achilles tendon reflex on the right. Sensory examination reveals diminished sensation over the lateral foot.
- ▶ **Question:**
What is the most likely diagnosis, and what are the recommended initial management steps?

► JAMA. Author manuscript; available in PMC: 2018 Mar 29.

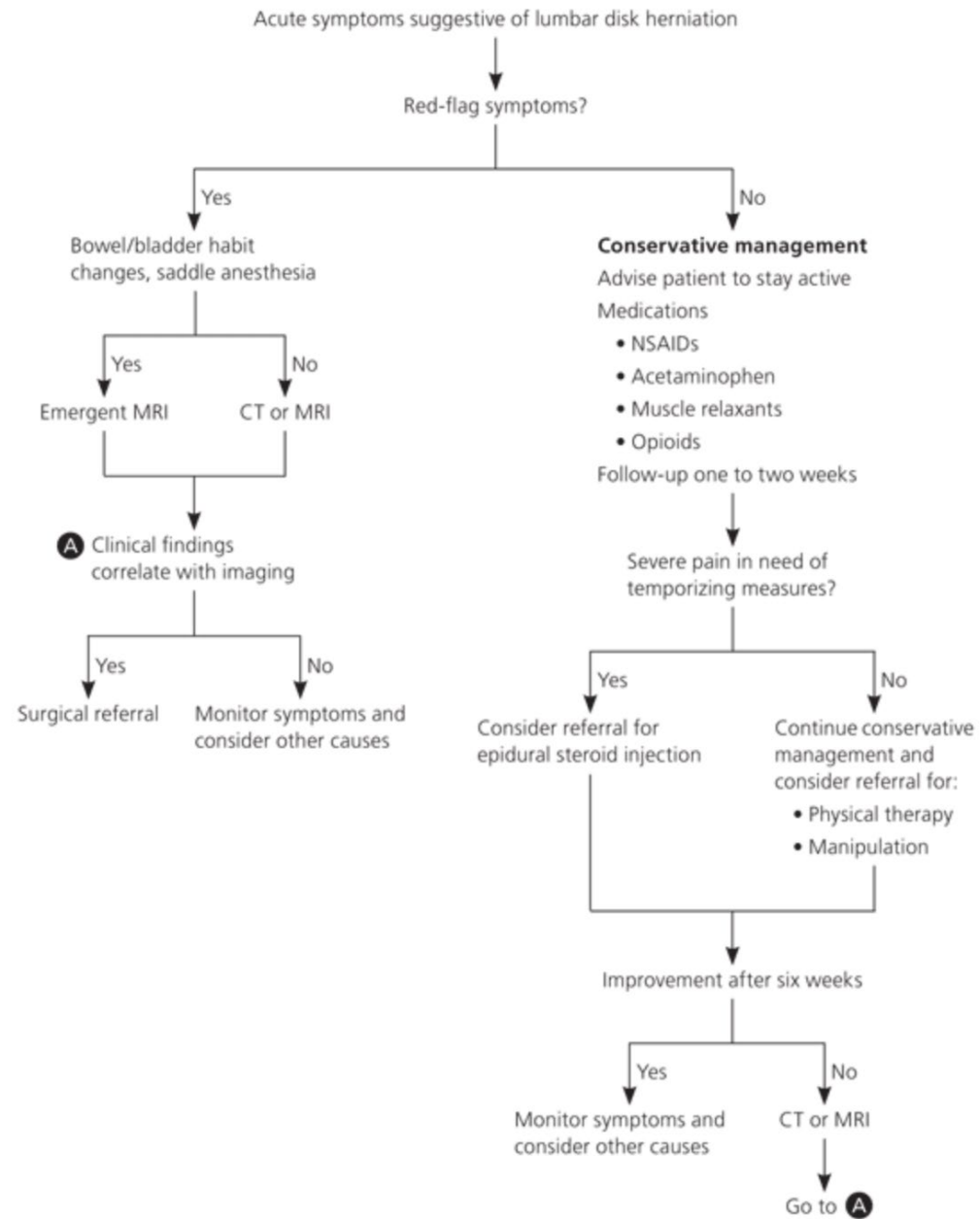
Published in final edited form as: JAMA. 2015 May 19;313(19):1915–1923. doi: [10.1001/jama.2015.4468](https://doi.org/10.1001/jama.2015.4468)

Oral Steroids for Acute Radiculopathy Due to a Herniated Lumbar Disk

A Randomized Clinical Trial

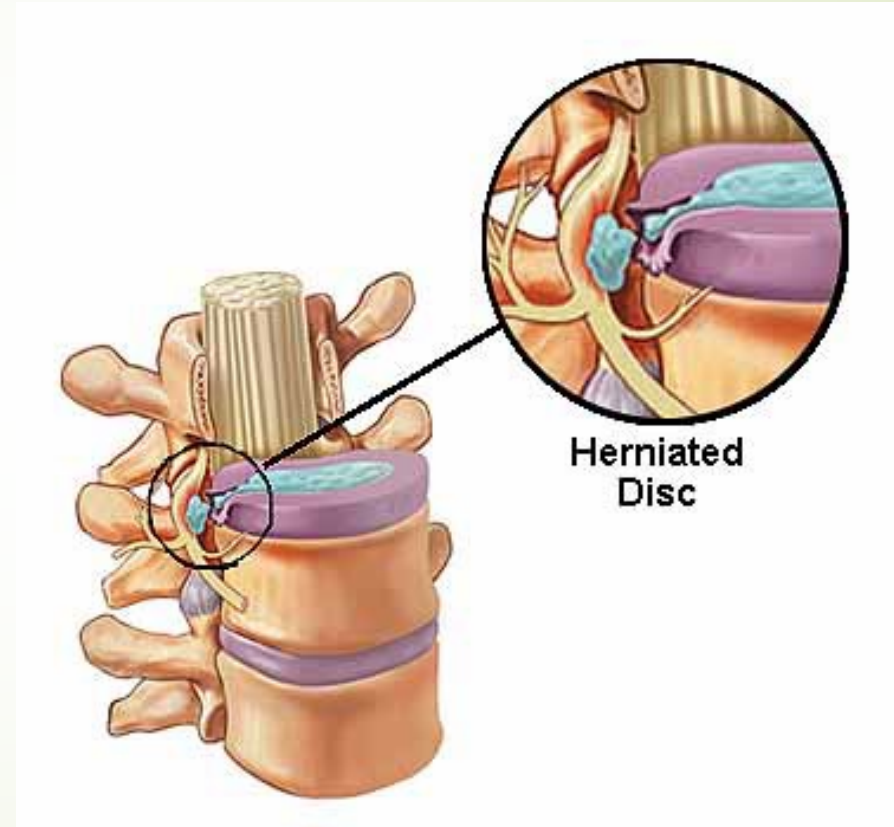
[Harley Goldberg](#)^{1,2}, [William Firth](#)³, [Mark Tyburski](#)⁴, [Alice Pressman](#)^{5,6}, [Lynn Ackerson](#)⁷, [Luisa Hamilton](#)⁸,
[Wayne Smith](#)⁹, [Ryan Carver](#)¹⁰, [Annu Maratukulam](#)¹¹, [Lawrence A Won](#)¹², [Eugene Carragee](#)¹³, [Andrew L Avins](#)
^{14,15,16}

- **Answer:**
The most likely diagnosis is **acute lumbar radiculopathy due to a disc herniation at the L5/S1 level**.
- Initial management includes conservative treatment with NSAIDs or acetaminophen for pain relief, physical therapy focused on strengthening and flexibility, and **activity modification*** to avoid aggravating positions.
- OMM: indirect techniques followed by gentle direct
- Oral steroid taper with improved functional status*
- Consider neuropathic agents such as Gabapentin and muscle relaxers.
- If no symptoms relief**, consider interventional procedures such as lumbar epidural steroid injections.
- If symptoms persist beyond 6-8 weeks or if there is progressive neurological deficit, imaging (MRI of the lumbar spine) and possibly referral to a spine specialist may be warranted.



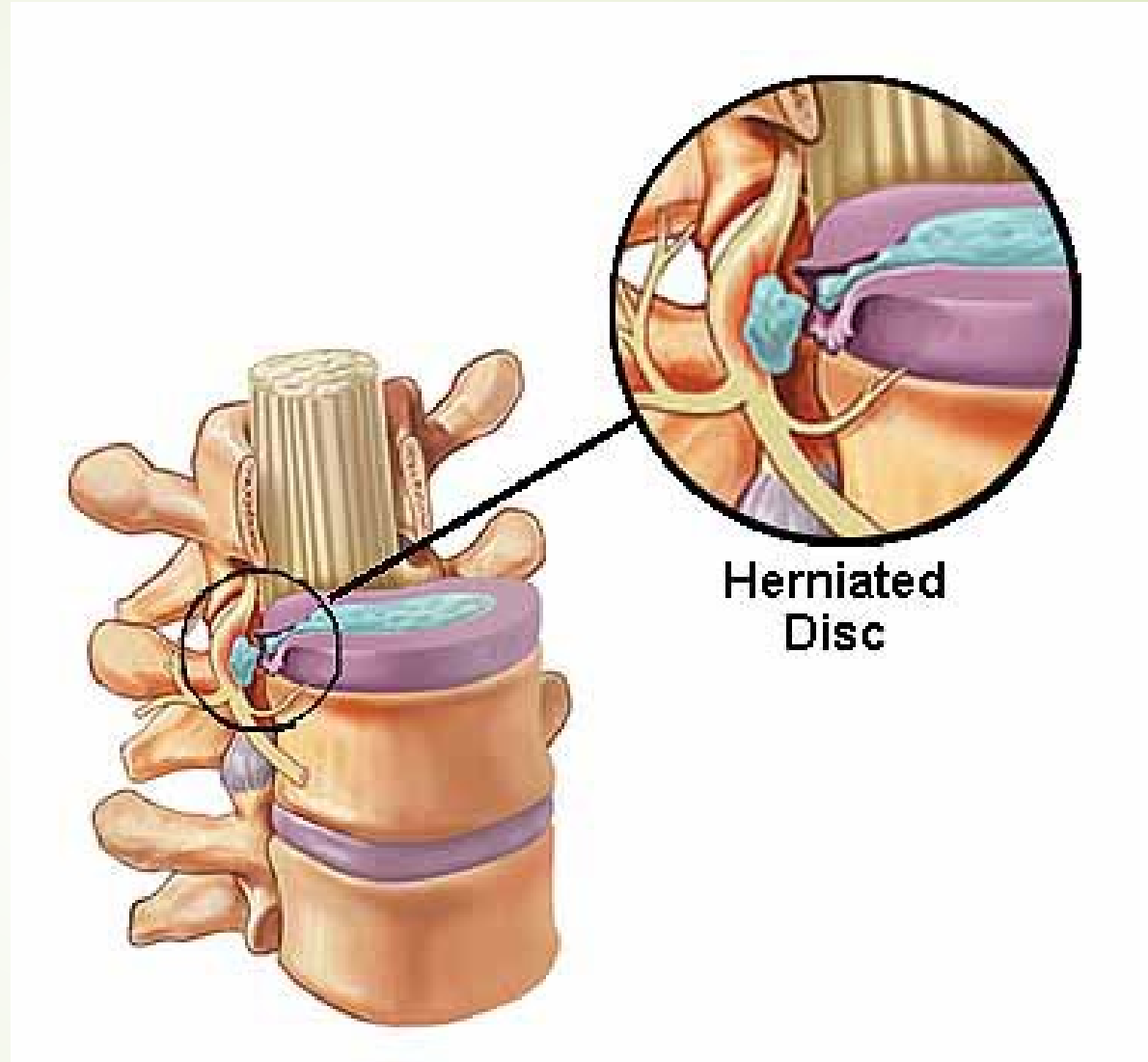
Development of Disc Herniation

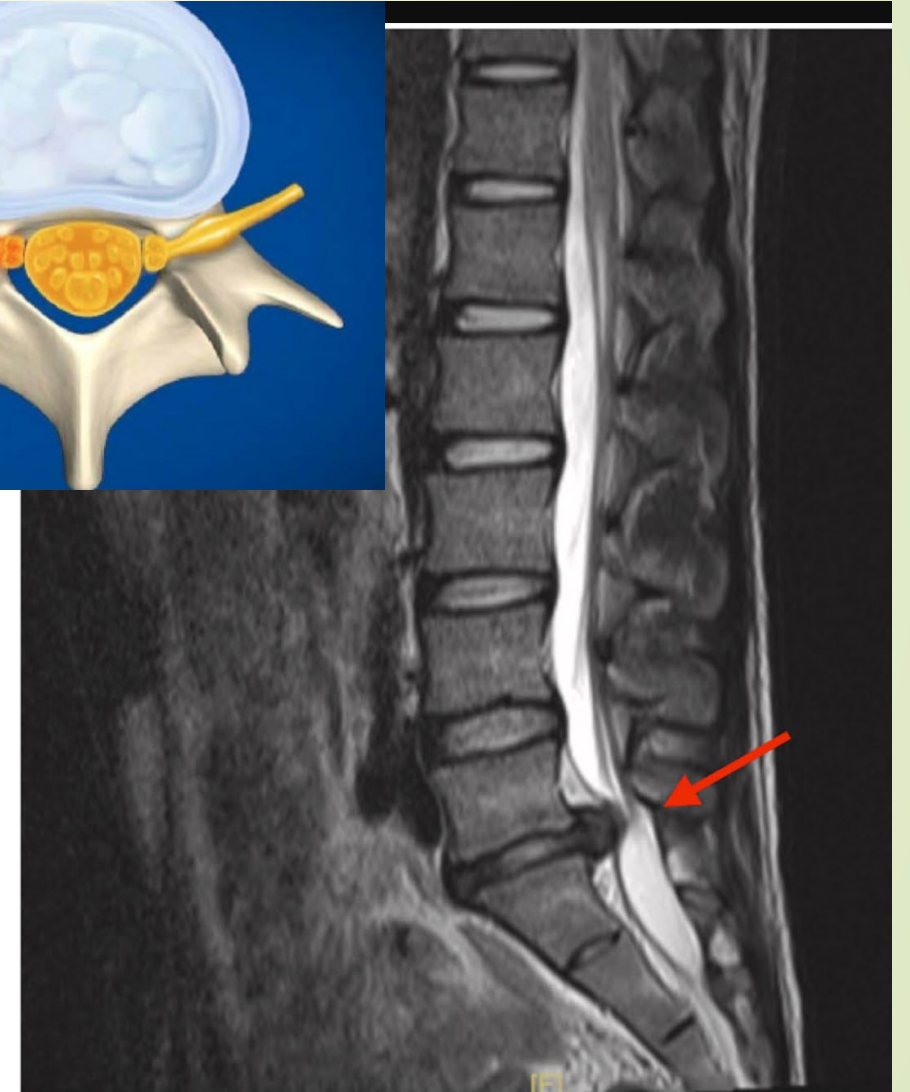
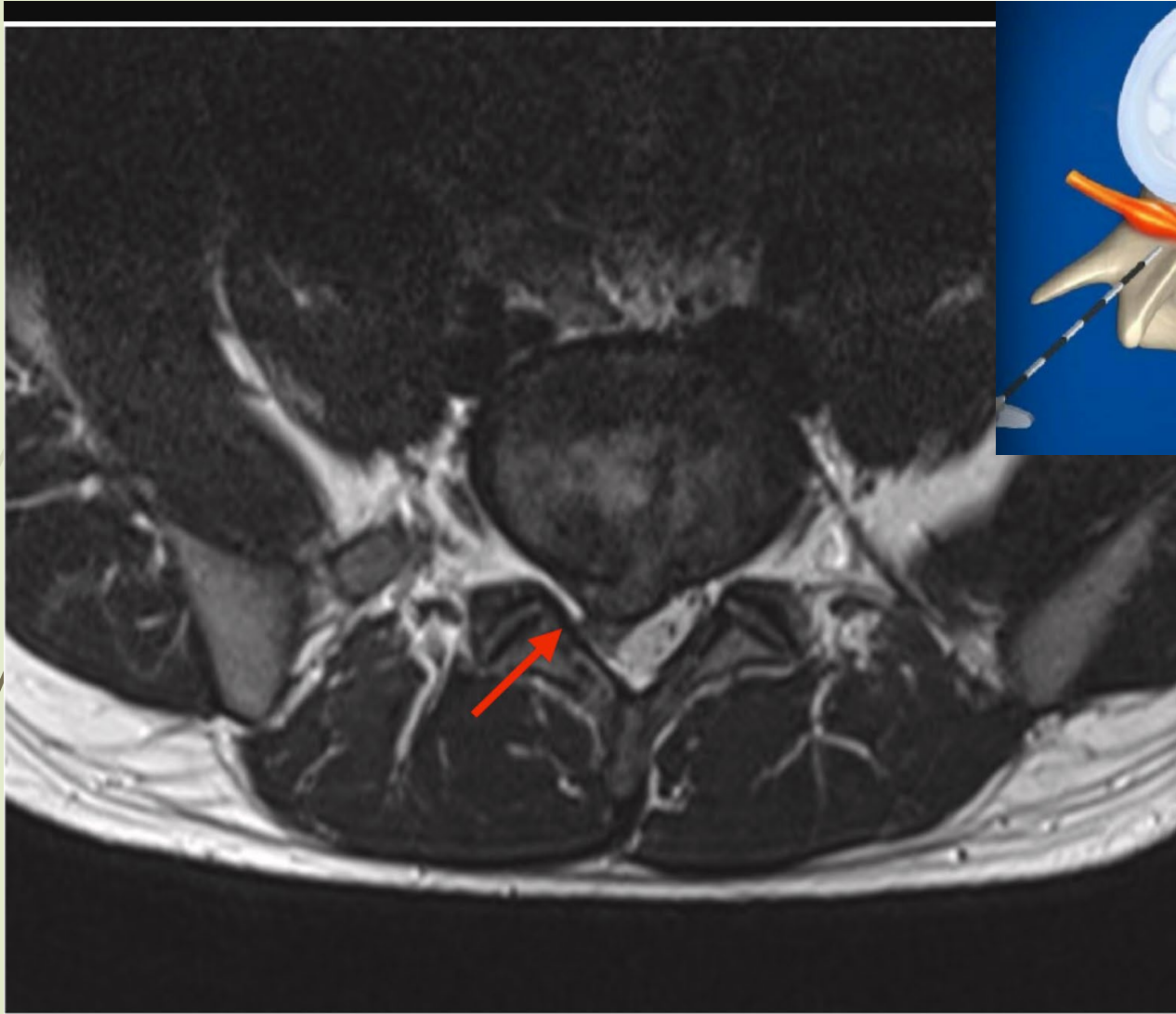
- Structural integrity of disc is reduced with degeneration
- Tears occur in the annulus
- Disc material can protrude thru the tear
- Displaced material can result in nerve compression



Pain and Disc Herniation

- Direct compression of the nerve root causes pain and neurological deficit
- Inflammatory reaction to disc nucleus results in nerve pain





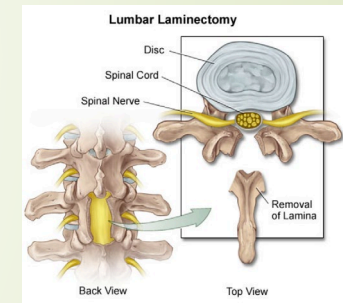
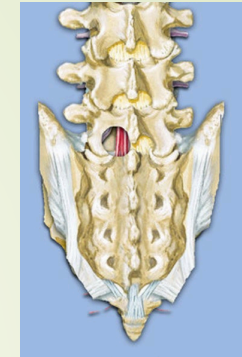
Lumbar Disc Syndromes Comparisons

- L4 radiculitis
 - Weak Anterior tibialis
 - Depressed knee reflex
 - Sensory loss big toe
- L5 radiculitis
 - Weak Extensor Hallucis
 - No reflex abnormality
 - Sensory loss mid-foot
- S1 radiculitis
 - Weak gastroc-soleus
 - Depressed ankle reflex
 - Sensory loss lat. foot

Nerve root	L4	L5	S1
Pain			
Numbness			
Motor weakness	Extension of quadriceps.	Dorsiflexion of great toe and foot.	Plantar flexion of great toe and foot.
Screening exam	Squat & rise.	Heel walking.	Walking on toes.
Reflexes	Knee jerk diminished.	None reliable.	Ankle jerk diminished.

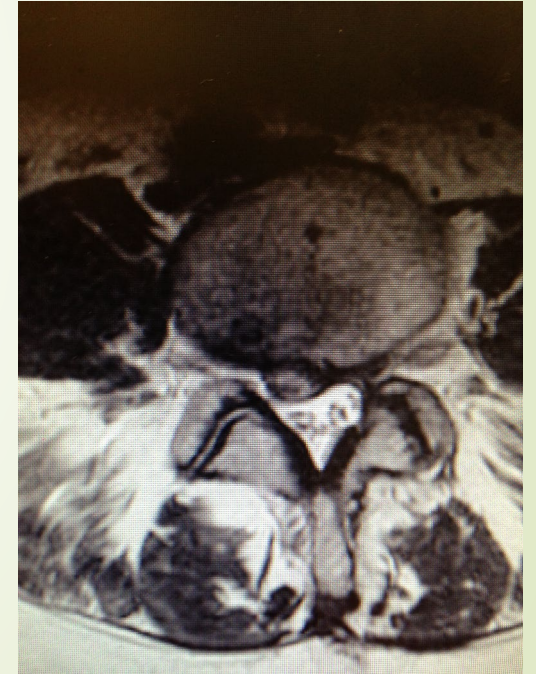
Spine surgery Terminology

- -otomy = cutting of
 - Osteotomy = cutting of bone
 - Laminotomy = removal of part of lamina to access spinal canal
- -ectomy = removal of
 - Laminectomy = removal of laminae
 - Vertebrectomy/corpectomy = removal of vertebral body
- Fixation = placing metallic implants between bones to prevent motion
- Fusion (arthrodesis) = fusing two bony structures together



Relative Indications for Lumbar Discectomy for radiculopathy

- Failure of 4-6 weeks non operative care
- Persistent radicular leg pain
- Neurological changes consistent with nerve root compression
- (+) SLR
- Confirmatory study identifying nerve root compression that correlates with clinical findings



Absolute Indications for Spinal Decompression

- Loss of bowel or bladder function
- Progressive neurological loss
- Profound neuro loss
- Intractable severe radicular pain



Clinical Case

- ▶ A 68-year-old woman presents to the clinic with complaints of lower back pain and progressive difficulty walking over the past several months. She reports that her pain radiates down both legs, particularly along the back and sides of her thighs and calves. The pain and numbness worsen with prolonged standing and walking, causing her to stop and rest frequently.
- ▶ She notes that leaning forward, such as when pushing a **shopping cart**, provides significant relief. She denies any recent trauma but mentions that her symptoms are gradually limiting her mobility. On physical examination, there is mild weakness in her lower extremities, but her reflexes are intact. Straight leg raise test is negative, and there is no evidence of acute nerve root tension.
- ▶ **Question:**
What is the most likely diagnosis, and what are the recommended initial management strategies?

➤ **Answer:**

The most likely diagnosis is **lumbar spinal stenosis**.

- Initial management includes patient education, physical therapy focused on flexion-based exercises to open the spinal canal at least 4-6 weeks. Oral medications: NSAIDs and neuropathic agents .
- OMM: decrease restrictions, improve range of motion.
- Interventional procedures such as lumbar epidural steroid injections should be considered if not significant relief. MRI should be ordered prior to this.
- Consider minimally invasive procedure before surgery*



Spinal Stenosis

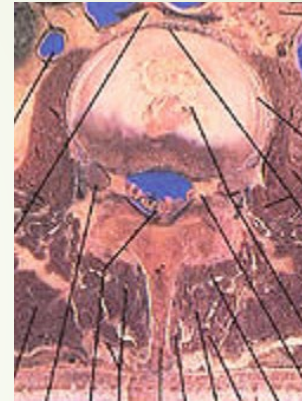


Definition

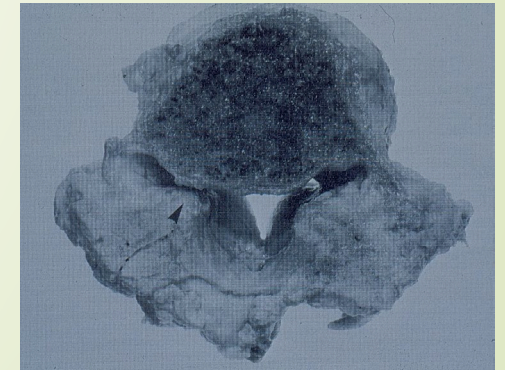
- Narrowing of the spinal canal
- Not necessarily symptomatic
- Slow onset compared to disc herniation which is more acute
- Despite significant compression nerve roots adapt to compression
- When need recruitment of motor units for activity (walking) may become symptomatic
- As activity terminated, after period of recovery (>5-10 min) symptoms remit

Spinal stenosis

- Claudication
 - Leg pain, fatigue with walking
 - Neurogenic
 - Slow recovery
 - Relief with flexion
- Rarely associated with radiculopathy
 - Insidious onset
- Neuro changes, Bladder dysfunction
- Common to have deformity such as s spondylolisthesis or lumbar scoliosis



normal

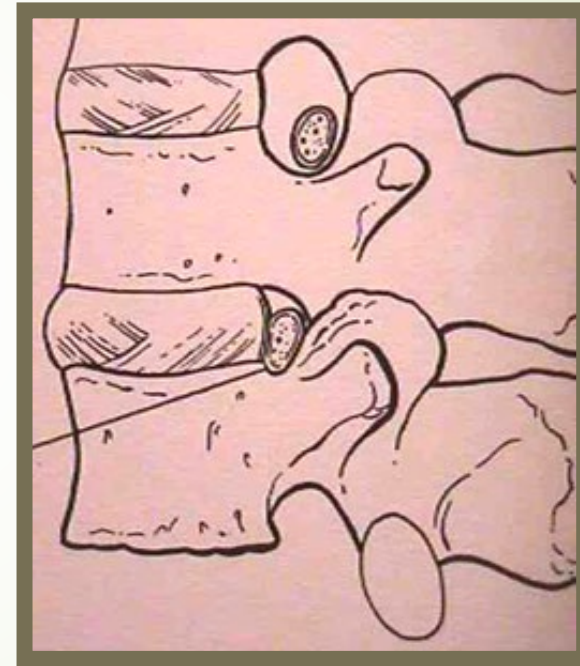


Stenosis

Stenosis



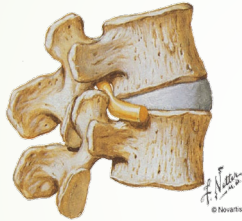
Central stenosis



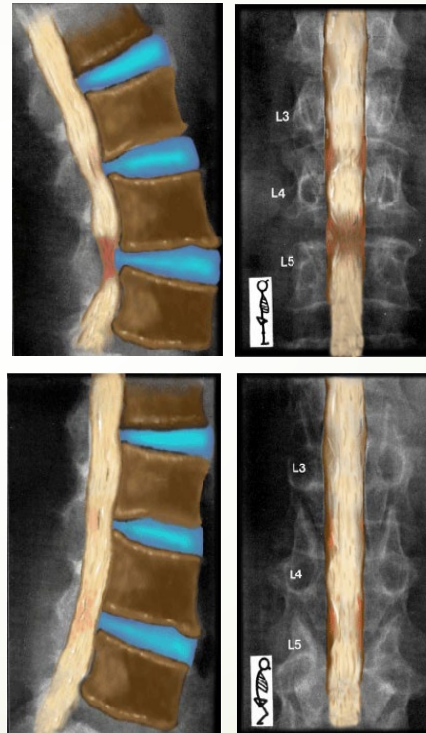
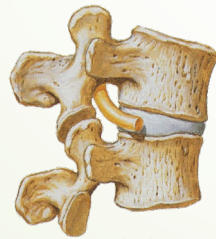
Foraminal Stenosis

Changes in the spinal Canal with Flexion and Extension

Spinal canal
Decreases
in Extension



Increase in
Flexion



Clinical Presentation of LSS



Extension provokes
symptoms of
pain/weakness in
the legs



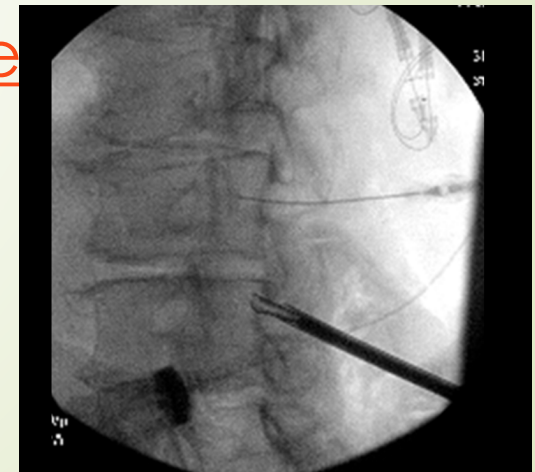
Patients lean forward
while walking to
ambulate more
comfortably



Sitting relieves
symptoms

Minimally Invasive Procedures

- Interspinous spacers:
- <https://youtu.be/jWok58xLM20>
- Watch starting 1:06
- Minimally Invasive Lumbar Decompression (MILD)
- [Mild Procedure Patient Education | Minimally Invasive Treatment](#)
- Watching starting 2:08



Surgical Treatment

Decompression

Laminectomy, Laminotomy

Foraminotomy/facetectomy

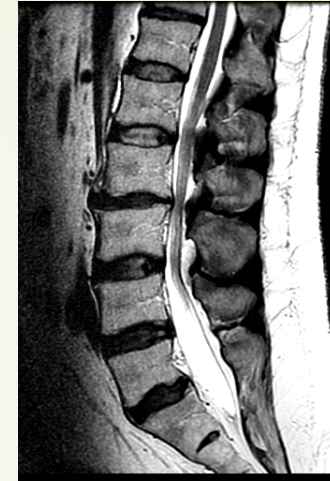
Fusion – If deformity present

Outcome

148 Pts with 10 yr follow up

	1 yr	4 yrs	10 yrs
Surgery (81 Pts)	77%	70%	54%
No surgery (67 Pts)	44%	52%	42%

Atlas SJ, Delitto A. Spinal stenosis: surgical versus nonsurgical treatment. *Clinical orthopaedics and related research*. Feb 2006;443:198-207.

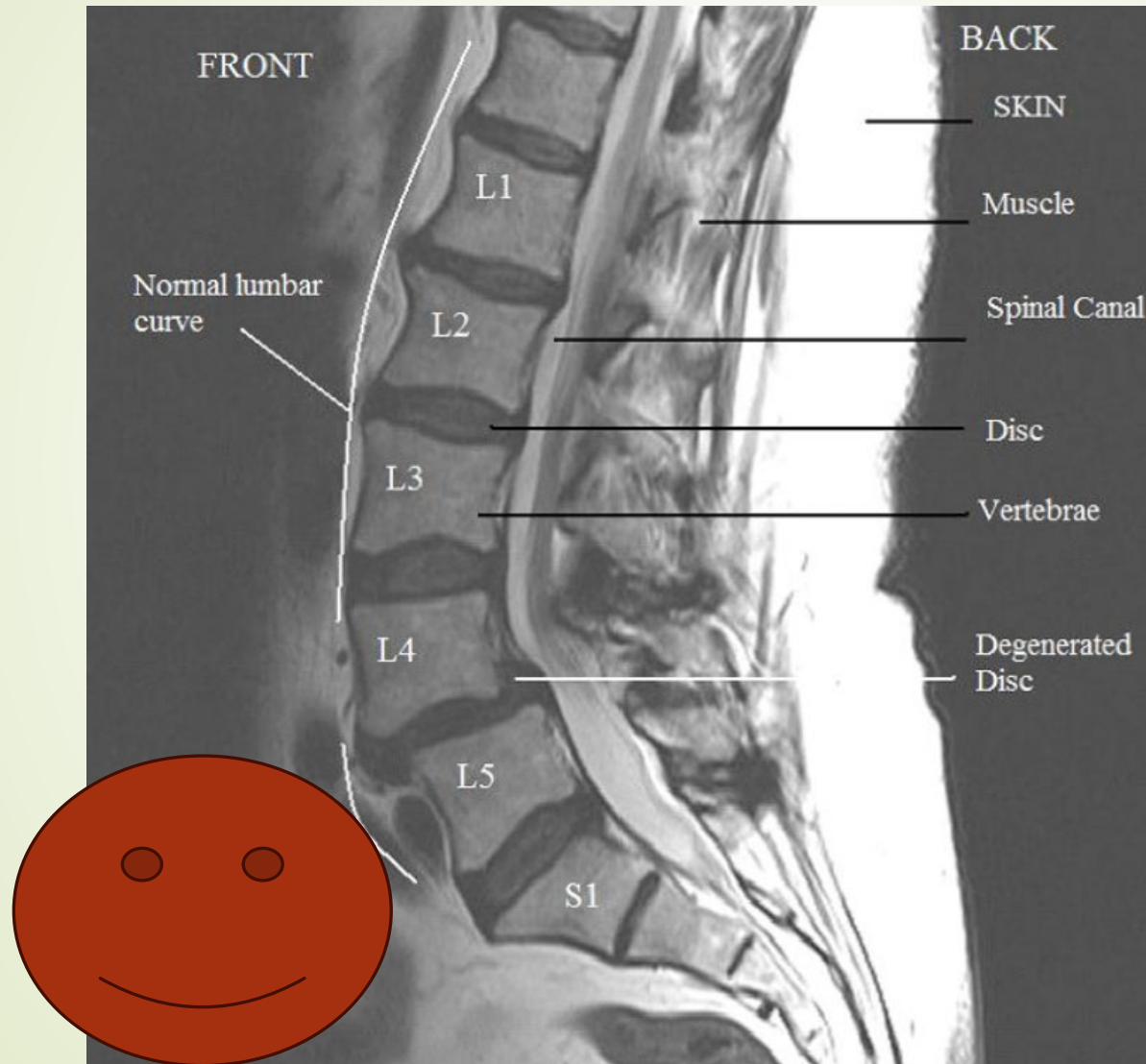


Clinical Case

- ▶ A 50-year-old woman presents to her primary care physician with complaints of chronic lower back pain that has gradually worsened over the past year. She describes the pain as a dull, aching discomfort in her lower back that sometimes radiates to her buttocks and upper thighs. She reports that the pain is worse when standing or walking for prolonged periods and is relieved by lying down. Recently, she has also noticed a sensation of weakness in her legs after walking long distances, which causes her to rest more frequently. She denies any recent trauma or significant weightlifting but mentions a history of gymnastics during her teenage years. On examination, there is tenderness over the lower lumbar spine, particularly at the L4-L5 level. Her lumbar lordosis appears exaggerated. Straight leg raise test is negative, and lower extremity strength and reflexes are normal. She brings an old lumbar spine x-ray.

- ▶ **Question:**
What is the most likely diagnosis, and what are the recommended initial management steps?

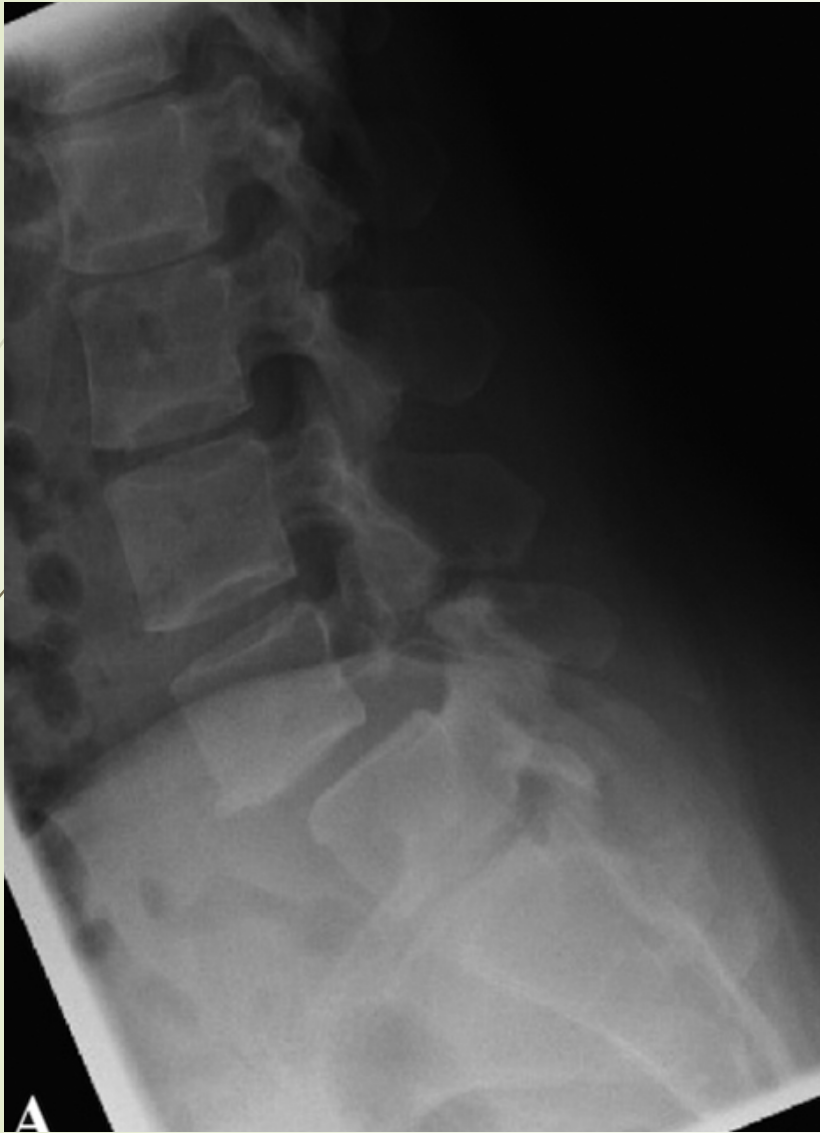
Old Lumbar Spine X-ray



➤ **Answer:**

The most likely diagnosis is a likely **grade 1-2 lumbar spondylolisthesis**, likely at the L4-L5 level.

- Initial management includes conservative treatment with physical therapy focusing on core strengthening and postural training to help stabilize the spine. NSAIDs or other analgesics can be used for pain relief. Activity modification, avoiding activities that exacerbate the pain, is also recommended.
- OMM: Reduce lumbar lordosis. Avoid HVLA
- If symptoms persist or worsen, imaging studies (such as X-ray* or MRI)
- Interventional procedures



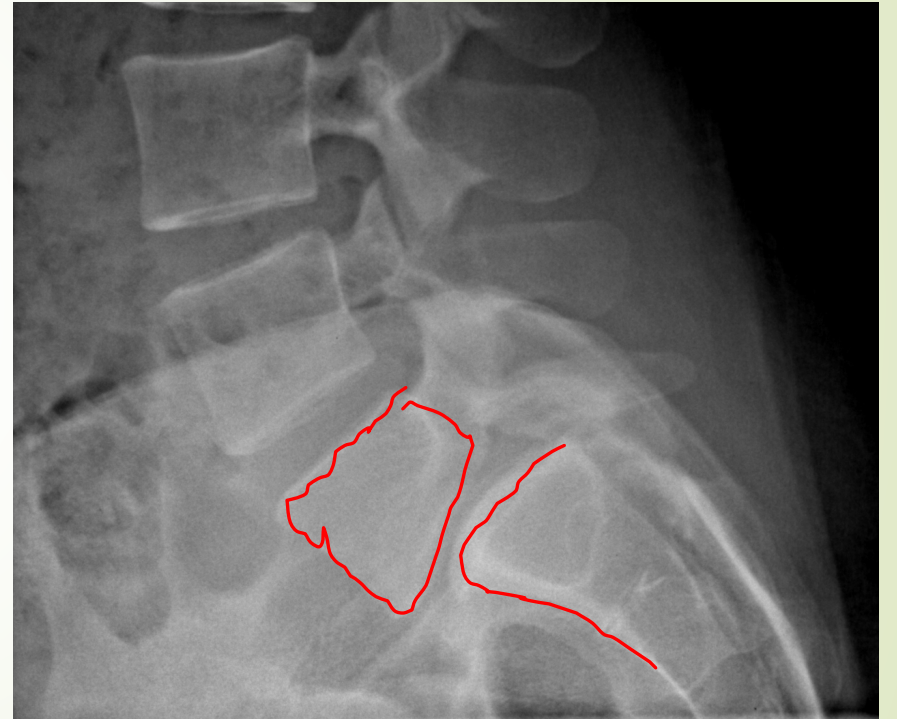
Spondylolisthesis

Definition

- “Spondy” = spine
- “Olisthesis” = slip
- Slip of one vertebral segment relative to another

4-6 % of adults have some form

Most are asymptomatic



Classification of spondylolisthesis

Spondylolisthesis:

Causes and Types

Degenerative:

- **Most common type:** Occurs in people over 50.
- **Causes:** Aging and wear-and-tear on the discs between vertebrae.
- **Related conditions:** Degenerative disc disease and osteoporosis.
- **Symptoms:** Back pain, leg pain, numbness, tingling.

Isthmic:

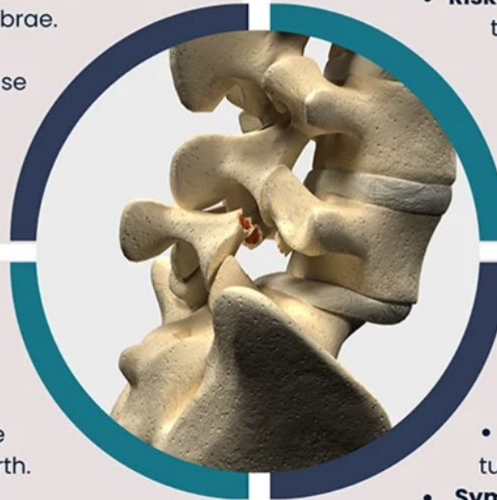
- **Causes:** Stress fractures in the vertebrae (spondylolysis).
- **Risk factors:** Athletes, especially those who hyperextend their backs (e.g., football, gymnastics).
- **Symptoms:** Back pain, leg pain, numbness, tingling.

Congenital:

- **Causes:** Abnormal bone formation present at birth.
- **Symptoms:** Back pain, leg pain, numbness, tingling.

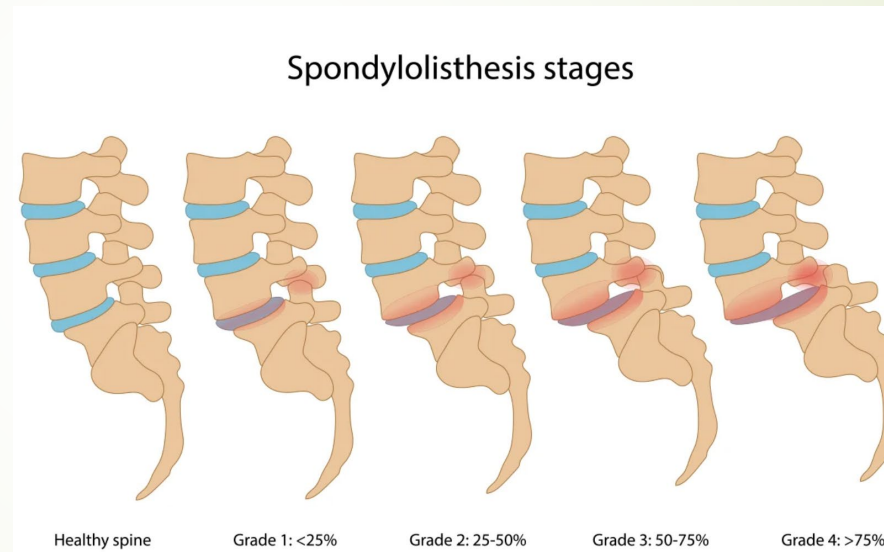
Other:

- **Causes:** Disease (infection, tumor), poor posture, obesity.
- **Symptoms:** Back pain, leg pain, numbness, tingling.



Severity of Spondylolisthesis is determined by the amount of slippage

Grade	% Slip
I	<25%
II	26 - 50%
III	51- 75%
IV	75% - Complete
Spondyloptosis	>100%



In younger patients, high grade (>50%) have higher chance of progressing and thus require surgery



Clinical Case

- A 55-year-old woman presents to the orthopedic clinic with severe lower back pain that has progressively worsened over the past several months. She describes the pain as a constant, aching sensation in her lower back, with intermittent sharp pains radiating down the back of her left leg to her calf and foot. She reports numbness and tingling in her left leg and difficulty walking more than a few minutes due to leg weakness and instability. She also mentions that she feels off-balance and often needs to sit or lean forward to relieve the discomfort. Her medical history is significant for chronic low back pain and mild spondylolisthesis diagnosed several years ago.
- On physical examination, there is an exaggerated lumbar lordosis and a noticeable step-off deformity at the lower lumbar spine. Straight leg raise test is positive on the left side, reproducing her radicular symptoms. There is mild weakness in dorsiflexion of the left foot and decreased sensation along the L5 dermatome. Reflexes are normal, but she has difficulty maintaining an upright posture without support. She brings in a previous lumbar spine MRI.
- **Question:**
What is the most likely diagnosis, and what are the recommended management steps?



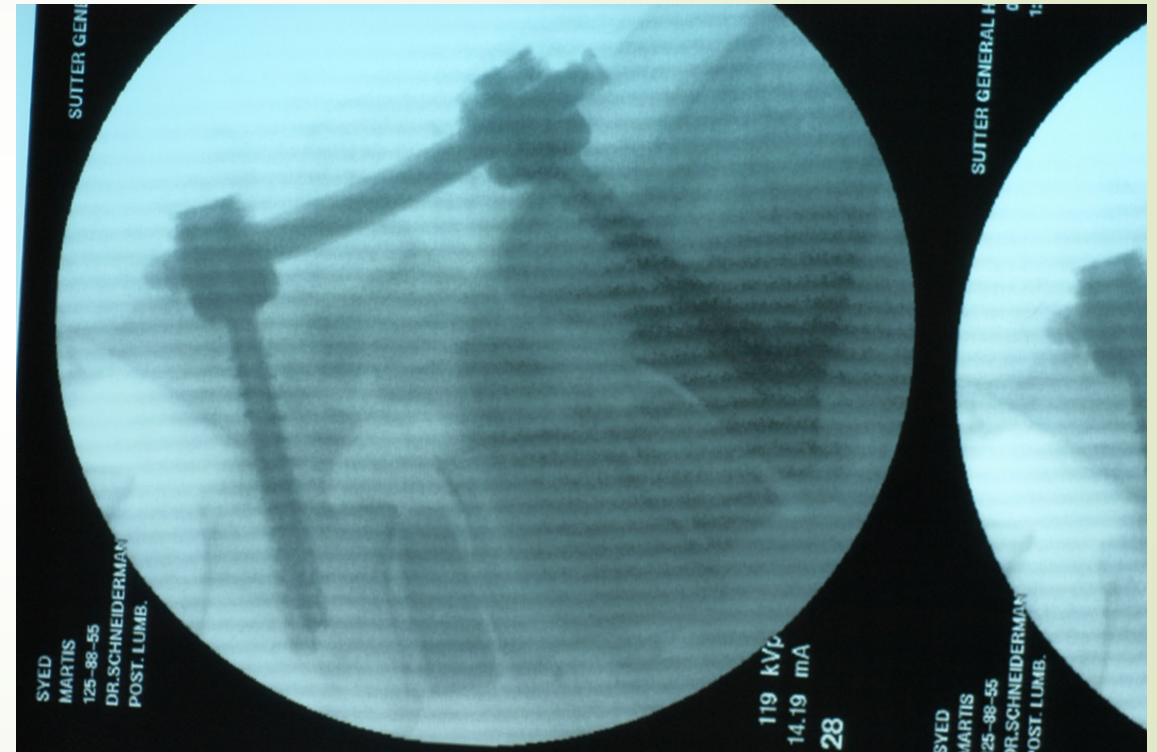
➤ **Answer:**

The most likely diagnosis is **grade IV lumbar spondylolisthesis** at L5-S1 level, with significant slippage causing nerve root compression and radicular symptoms.

- Initial management may involve conservative measures, including physical therapy and NSAIDs for pain relief.
- However, given the severity of symptoms, instability, and neurological involvement, imaging (such as an MRI or CT scan) is warranted to assess the degree of slippage and nerve compression.
- Referral to a spine surgeon is recommended for consideration of surgical decompression and spinal fusion to stabilize the spine and relieve nerve compression.

Treatment

- Anterior fusion and reduction of spondylolisthesis performed at L4-5
- Fusion anterior at L5-S1 due to adjacent degeneration
- Posterior Laminectomy to decompress the spinal canal and nerve roots
- Posterior instrumentation to stabilize until bone healing occurs at 6-18 months





Lumbar Facet Arthropathy	Degeneration of lumbar facet joints due to wear and tear	Localized lower back pain, worsens with extension and rotation; morning stiffness, worse with prolonged standing	Clinical diagnosis, MRI if needed	NSAIDs, PT focusing on core strengthening, lumbar stabilization exercises, facet joint injections if refractory
Lumbar Spinal Stenosis	Narrowing of spinal canal, causing nerve root compression	Leg pain with walking, relieved by sitting	MRI for confirmation	PT, NSAIDs, surgery if conservative measures fail
Lumbar Disc Herniation	Nucleus pulposus herniates through annulus fibrosus	Radicular leg pain, positive straight leg test	MRI if persistent symptoms	NSAIDs, PT, surgery if neurological symptoms worsen
Lumbar Spondylolisthesis	Anterior slippage of one vertebra over another	Back pain, possible radicular symptoms	X-ray to assess slippage	PT, NSAIDs, surgery for high-grade or symptomatic cases



Surgery did not resolve symptoms now what: **Failed Back Surgery Syndrome**

- Failed back surgery syndrome (FBSS) is defined by the International Association for the Study of Pain as lumbar spinal pain of unknown origin either persisting despite surgical intervention or appearing after surgical intervention for spinal pain originally in the same topographical location.
- Treatment:
 - Activity modifications, PT, Medications: NSAIDS/neuropathic agents,
- - Interventional Procedures
- - Neuromodulation
- - Revision Surgery



Clinical Case

- ▶ A 47-year-old man presents to the emergency department with sudden onset of severe lower back pain that began two days ago and has progressively worsened. He reports that the pain radiates down both legs, especially on the right, and is accompanied by numbness in his inner thighs and perineal area. He mentions that he has recently developed difficulty initiating urination and feels as though he cannot fully empty his bladder. Additionally, he notes weakness in his legs and has stumbled several times while walking. He denies any recent trauma or prior episodes of similar symptoms.
- ▶ On physical examination, there is decreased sensation in a "saddle" distribution over the perineal area, along with diminished sensation in the posterior legs. Motor examination reveals weakness in ankle dorsiflexion and plantarflexion on both sides. Reflexes in the lower extremities are reduced, and anal sphincter tone is diminished.

➤ **Question:**

What is the most likely diagnosis, and what is the recommended next step in management?

➤ **Answer:**

The most likely diagnosis is **cauda equina syndrome**, likely due to significant lumbar disc herniation or another compressive lesion. The next step in management is to urgently obtain an MRI of the lumbar spine to confirm the diagnosis and assess the level of compression.

- This is a surgical emergency, and immediate referral to neurosurgery for decompression is necessary to prevent permanent neurological deficits.



Thank You

